THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL UNIT TEST SET 3

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A longer wire of the same material and thickness has 1

(a) more resistance **(b)** less resistance
(c) no resistance **(d)** the same resistance

2. When the current in a wire is reversed, the compass needle deflects 1

(a) in the opposite direction **(b)** randomly
(c) in the same direction **(d)** not at all

3. A concave lens is thicker at the 1

(a) centre **(b)** only at one side **(c)** both equally **(d)** edges

4. Which colour has the shortest wavelength in the visible spectrum? 1

(a) Violet **(b)** Red **(c)** Green **(d)** Yellow

5. In series, if one bulb fuses, the others 1

(a) stop glowing **(b)** glow brighter **(c)** glow dimmer **(d)** are unaffected

6. In Fleming's left-hand rule, the thumb indicates 1

(a) force or motion **(b)** resistance **(c)** field **(d)** current

7. The perpendicular drawn at the point of incidence is the 1

(a) pole **(b)** axis **(c)** tangent **(d)** normal

8. The black opening in the middle of the iris is the 1

(a) pupil **(b)** cornea **(c)** retina **(d)** lens

9. The instrument used to measure potential difference is the 1

(a) barometer **(b)** ammeter **(c)** ohmmeter only **(d)** voltmeter

10. Unlike poles of magnets 1

(a) attract each other **(b)** melt
(c) repel each other **(d)** do not interact

11. The speed of light in vacuum is 1

(a) 3×10^5 m/s **(b)** 3×10^8 m/s **(c)** 3×10^{10} m/s **(d)** 3×10^6 m/s

12. Spectacles for hypermetropia use 1

(a) prisms **(b)** plane lenses **(c)** concave lenses **(d)** convex lenses

13. Resistance of a metal wire 1

(a) is the same at all temperatures **(b)** increases with temperature
(c) decreases with temperature **(d)** becomes zero on heating

14. The field lines around a straight wire are 1

(a) radial lines **(b)** parallel lines
(c) spirals only **(d)** concentric circles

15. The centre of a lens is called the 1

(a) pole **(b)** focus **(c)** aperture **(d)** optical centre

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is the formula for the equivalent resistance in series? 2

17. Who discovered electromagnetic induction? 2

18. State the nature of the image formed by a plane mirror. 2

19. Which part of the eye controls the size of the pupil? 2

20. What is the SI unit of charge? Give its value in terms of electrons. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A current of 0.2 A flows through a resistor of 10 ohm. Find the potential difference. 3

22. State any three properties of magnetic field lines. 3

23. Why does a ray of light bend when it enters water? 3

24. Define dispersion. Which colours are deviated the most and the least? 3

OR

Find the resistance of a series connection of 3 ohm, 4 ohm and 5 ohm. 3

25. State Fleming's right-hand rule. 3

OR

What is meant by the term 'principal axis'? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain why convex mirrors are used as rear-view mirrors. 5

OR

Explain how the human eye forms an image. Why is the image inverted but we see objects erect? 5

27. Explain the factors on which the resistance of a conductor depends. 5

OR

Describe Oersted's experiment with a diagram description. State its conclusion. 5

END OF QUESTION PAPER Total: 50 Marks